MODULE: CLASSIFICATION

Hierarchy of life forms at phenomenological level:

Paul Anderson explains how Biology is ordered in the hierarchy of life. The hierarchy is –

Atom
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Molecule
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Macromolecule
↓
Organelle
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Cell
↓
Tissue
↓
Organ
↓
Organ system
↓
Organism
↓
Population
↓
Community
↓
Ecosystem
↓
Biome
↓
Biosphere

Atom: Smallest part of a chemical that can take part in a chemical reaction. E.g. Carbon, Hydrogen, Nitrogen, Oxygen, etc.
Molecule: A group of atom. The smallest unit into which a substance can be divided without a change in its chemical nature. E.g. Glucose, Amino acid, etc.
Macromolecule: Large sized structure, generally formed by polymerisation of simple molecule. Has high molecular weight ranging 10000 Dalton and above. E.g. Protein, Starch, etc.
Organelle: Biomacromolecules like carbohydrate, protein, etc. interact to form organelles, which are membrane bound intracellular structures performing specialized tasks. E.g. Endoplasmic reticulum, Lysosome, Golgi apparatus, etc.
Cell: Structural and functional unit of life. Prokaryotic cell, Eukaryotic cell.
Tissue: A group of similar or dissimilar cells, having a common origin and performing similar function. E.g. Epithelial tissue, Meristematic tissue, etc.
Organ: Many tissue become engaged to form a particular organ, performing a specialized task or function. E.g. Liver, Testes, etc.
Organ system: It emerges when several organs take up a particular life process. E.g. Digestive system, Excretory system, etc.
Organism: The living body or individual. Human, Mango plant, etc.

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Population: A localised group of organisms belonging to the same species constitute a population. E.g. Human population, Cat population, etc.
Community: Populations of different species living in the same area make up a biological community. E.g. Desert community, Forest community, etc.
Ecosystem: Community interactions, integrated with the abiotic features of the environment form an ecosystem. E.g. Desert ecosystem, Grassland ecosystem, etc.
Biome: It is a large community unit established by a complex interaction of climate, other physical factors and biotic factors. E.g. Forest biome, Alpine biome, etc.
Biosphere: The entire zone of air, land and water, at the surface of the earth that is occupied by organisms.

Underlying criteria for classification:

Biological classification is the process by which living organisms are grouped into easy identifiable groups or categories based on some observable characters. The underlying criteria for classification can be –

1. **Morphology** – It deals with the study of shape, size, colour, texture and structure of the organism.
2. **Biochemistry** – It deals with the study of chemical constituents of organisms which are deeply concerned with the molecular characteristics.
3. **Ecology** – It deals with the study of distribution, structural adaptation, other various aspects of life of organisms and their interactions with other organisms and environment.

Discussion of classification on the basis of different parameters:

1. **Cellularity:** On the basis of cellularity, organisms can be classified into two types. If the organism is made up of one cell only then it is unicellular, for example, amoeba, yeast, euglena, etc. However, if the organism is made up of many cells then it is called multicellular, for example, cat, human, mango plant, etc.
2. **Ultrastructure:** On the basis of cellular ultrastructure, organisms can be classified into two types. If the organism possesses cells with unorganised nucleus, then they are called prokaryotic organism, for example, bacteria, cyanobacteria, mycoplasma, etc. However, if the organism possesses cells with organised nucleus, then they are called eukaryotic organism, for example, amoeba, mosses, banyan tree, human, etc. Enlisted below is the difference between prokaryotes and eukaryotes.

Prokaryotes	Eukaryotes
Size mostly 1 to 10 µm.	Size mostly 10 to 100 µm.
Nucleus absent.	Nucleus present.
Genetic material circular, double stranded DNA.	Genetic material linear, double stranded DNA.
Membrane bound cell organelle absent.	Membrane bound cell organelle present.
Ribosome 70S type.	Ribosome 80S type.
Does not exhibit division of labour.	Exhibit division of labour.

3. **Energy and Carbon utilisation (Nutrition):** On the basis of mode of nutrition, organisms can be classified into three types. If the organism can manufacture their own food from simple inorganic raw materials present in the surrounding, then they are called autotrophs, for example, green plants, photosynthetic bacteria, cyanobacteria, photosynthetic protozoan, etc. If the organism cannot prepare their

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food by themselves and are directly or indirectly dependent upon autotrophs for their food then they are called heterotrophs, for example, human, cow, tiger, etc. However, there are another group of organisms that use inorganic compounds as electron donor to conserve energy for growth and organic carbon or CO_2 as a source of carbon, they are called lithotrophs, for example, sulphur bacteria, etc.

4. **Nitrogenous waste excretion:** On the basis of nitrogenous waste excretion, organisms can be classified into four types. If the organism excretes ammonia, then it is called ammonotelic for example, protozoa, etc. If the organism excretes urea, then it is called ureotelic, for example, mammals, etc. If the organism excretes uric acid, then it is called uricotelic, for example, insects, etc. If the organism excretes amino acid, then it is called aminotelic, for example, some amphibians, etc.
5. **Habitat:** On the basis of habitat, organisms can be classified into two types. If the organism live in water bodies, then they are called aquatic, for example, fish, etc. If the organism live on land, then they are called terrestrial, for example, horse, etc.

Molecular taxonomy – three major kingdoms of life (Three Domains of Life):

Concept of three domains of life was proposed by Carl Woese *et.al.* It presents an evolutionary model of classification, based on –

- Difference in the sequence of nucleotides in the cell's ribosomal RNA (rRNA).
- Lipid structure of cell membrane.
- Sensitivity to antibiotics.

Modern concept of biological classification has evolutionary basis. Since rRNA molecules have most conserved structure throughout the nature and perform the same function, similarities and differences in rRNA nucleotide sequence are a good indication of relationship in different living groups.

It is now accepted that three distinct domains of organisation evolved in nature from a common archaeal-prokaryotic ancestor cell. These Domains are –

1. **Domain Archae:** Members of this domain have cell membrane composed of branched hydrocarbon chains attached to glycerol. Their cell wall has no peptidoglycans, rather it may be a glycoprotein or pseudomurein or chondroitin sulfate like substance or proteinaceous sheath. They are not influenced by antibiotics that destroy bacteria. E.g. Methanogens, Halophiles, Hyperthermophiles, etc.
2. **Domain Eubacteria:** Members of this domain have cell membrane composed of unbranched fatty acid chains attached to glycerol. Their cell wall is formed of peptidoglycans. They are sensitive to traditional antibiotics. E.g. Bacteria, Cyanobacteria, Mycoplasma, etc.
3. **Domain Eukarya:** Members of this domain have cell membrane composed of trilaminar protein-lipid-protein layer as in bacteria. The cell wall is absent in animals cells, but present in plant and fungal cell. Plant cell wall is mainly made up of cellulose, while it is chitin made in fungi. They are resistant to traditional antibiotics. E.g. Plants, Animals, Fungi and Protista.

A given organism can come under different category based on classification (Taxonomical Hierarchy):

Classification is a multi-step process. It involves hierarchy of steps and each step represents a rank or category. Each taxonomic category is a unit of classification and is termed as taxon.

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Taxonomic hierarchy is the arrangement of taxonomic groups or taxons in a definite order from higher to lower categories. The categories used in classification of animals are Kingdom, Phylum, Class, Order, Family, Genus and Species. In case of plants Phylum is replaced by Division. Thus in taxonomic hierarchy, there are minimum seven categories, called taxa. The Species is the lowest taxonomic category and the kingdom the highest category. Linnaeus introduced the hierarchical system of classification.

1. **Species:** A group of organisms that closely resemble freely interbreed and produce fertile offsprings.
2. **Genus:** It is the next category higher to the species. It comprises a group of related species which have more characteristics in common than the species of other genera.
3. **Family:** It is a group of related genera with still less number of similarities as compared to genus and species.
4. **Order:** The next higher category is the order which includes related families.
5. **Class:** The related orders are included in a class. The similarities in the orders of a class are still less than in the families of an order.
6. **Phylum / Division:** it includes different classes having very few similarities.
7. **Kingdom:** The highest category for a given organism. For example, all animals belonging to various phyla are placed together in the kingdom Animalia and all plant divisions are put together in kingdom Plantae.

Study of model organisms:

1. ***Escherichia coli:*** This bacterium has long been the favoured organism for investigation of the basic mechanisms of molecular genetics. Most of our present concepts of molecular biology, including understanding DNA replication, genetic code, gene expression and protein synthesis derive from studying *Escherichia coli*. It is especially useful to molecular biologists because of:
    - Relative simplicity.
    - Ease with which it can be propagated and studied in laboratory.
    - Small size of its genome (approximately 4300 genes).
    - Rapid growth under well-defined laboratory condition.

2. ***Saccharomyces cerevisiae*:** Yeast is one of the simplest eukaryotic organisms, but has several essential cellular processes in common with human, making it an important organism to study and understand basic molecular processes in human. It is a model organism because:
    - In laboratory it is easy to manipulate, and can cope with a wide range of environmental conditions.
    - Controls cell division in a similar way to human cells.
    - At least 20% of human genes known to have a role in disease have functional equivalents in yeast.

3. ***Drosophila melanogaster:*** Fruit fly has been a crucial model organism in developmental biology and genetics, due to the following features:
    - Moderate size of its genome (approximately 14000 genes).
    - It can be easily maintained and bred in the laboratory.
    - Short reproductive cycle (about 2 weeks).

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Many fundamental concepts of genetics, such as relationship between gene and chromosome, were derived from studies on fruit fly. Genetic analysis of *Drosophila melanogaster* has uncovered many genes that control development and differentiation.

4. ***Caenorhabditis elegans:*** This nematode worm is a model for research for the following reasons:
    - They can be grown cheaply and in large numbers as they can produce over 1000 eggs per day.
    - Healthy cultures can be frozen and then defrosted and revived when needed.
    - The worm is transparent throughout its life, so the behaviour of individual cells can be followed through its development.
    - Though relatively simple organism, many of the molecular signals controlling its development are also found in complex organism like human.
    - Many genes in its genome have functional counterparts in human which make it an extremely useful model for human diseases.

5. ***Arabidopsis thaliana:*** It is a small dicotyledonous species, a member of Brassicaceae or mustard family. Several traits that make it very desirable for laboratory study are:
    - It has a fast life cycle.
    - It produces numerous self-progeny.
    - It has very limited space requirements and is easily grown in a green house or indoor growth chamber.
    - It possesses a relatively small, genetically tractable genome that can be manipulated through genetic engineering more easily and rapidly than any other plant genome.

6. ***Mus musculus:*** Over the past century mouse has become the preferred mammalian model for genetic research for the following reasons:
    - Mouse has many similarities to human in terms of anatomy, physiology and genetics.
    - Mouse genome is very similar to our own, making it useful for studying human genetic diseases.
    - They are cost effective.
    - Adult mice multiply quickly, can produce as often as every 3 weeks.
    - Generation time (time between being born and giving birth) short, usually 10 weeks.
    - It is relatively easy to manipulate mouse genome.

    Now, scientists use mouse to stimulate human genetic disorders in order to study their development and test new therapies.

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